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A triboelectric nanogenerator sensor based on phononic crystal structures for smart buildings and transportation systems

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ABSTRACT

Many buildings and equipment are frequently disturbed by various types of elastic wave or vibration sources in ambient environment. The real-time state monitoring of such a disturbance is very important in the fields of green buildings and intelligent transportation systems. Based on triboelectric nanogenerators (TENGs), we propose a two-dimensional phononic crystal TENG (PC-TENG), which can serve as a self-powered sensor. The PC-TENG system consists of a periodical spherical cavity type of plate-like structure and elastic balls placed in each spherical cavity. Particularly, with bandgap property of phononic crystals, the proposed PC-TENG can also absorb or suppress vibration impact with certain frequencies acting on buildings and equipment. Taking elasticity, diameter, and surface roughness of elastic ball into account, we established a theoretical model of the cavity-ball type of PC-TENG based on Hertzian contact model. The sensing performance of the proposed PC-TENG with 5 by 5 cavities are experimentally measured, which shows the PC-TENG can efficiently detect the amplitude and frequency of an elastic wave disturbance. The theoretical output charges and currents evaluated from the developed theoretical model agree well with the experimental data.

1. Introduction

Triboelectric nanogenerators (TENGs), originally proposed by professor Zhonglin Wang, can convert mechanical energy into electrical energy through contact electrification (or triboelectrification) [1–3]. The TENGs are used as not only harvesters collecting mechanical energy in the surrounding environment, but also smart sensors monitoring working state of some equipment. The performance of TENG-based devices is mainly dependent upon the triboelectric effect and electrostatic induction. For TENGs, there are four kinds of common-used operating modes, namely, contact-separation, lateral sliding, single-electrode, and freestanding triboelectric-layer modes [4]. TENGs have been used to collect mechanical energy in different forms, such as mechanical movement of humans [5,6], wind [7,8] and water [9–11], existing in the environment. TENGs have been used for self-powered sensors [12–20] as well.

Mechanical vibrations with different frequencies [21,22] happen all the time in surrounding environment such as buildings, rails, and bridges etc., as shown in Fig. 1(a). For these vibrations, TENGs can be used to monitor their vibration states and collect their mechanical energy and show a good sensing and harvesting performance [23]. Researchers usually fabricated TENGs with multilayered (or other complex) structures for enhancing the energy output performance of TENGs. For example, a triple-cantilever-based TENG, which can produce an open-circuit voltage up to 101 V and a short-circuit current of 55.7 μA with a peak power density of 252.3 mW/m^2, is put forward in Ref. [24]; a multi-layered structure-based TENG [25–27] and a spring-based TENG [28] are proposed. As self-powered active sensors, TENGs can detect frequency, amplitude and position of vibration sources [29–32]. For instance, Wang et al. [33] developed a TENG with freestanding contact mode to quantitatively measure the vibration amplitude, Zhang et al. [34] proposed a self-powered distributed water level sensor. In combination with piezoelectricity and electromagnetic [35–40] coupling through coil, some researchers proposed hybrid TENGs with higher

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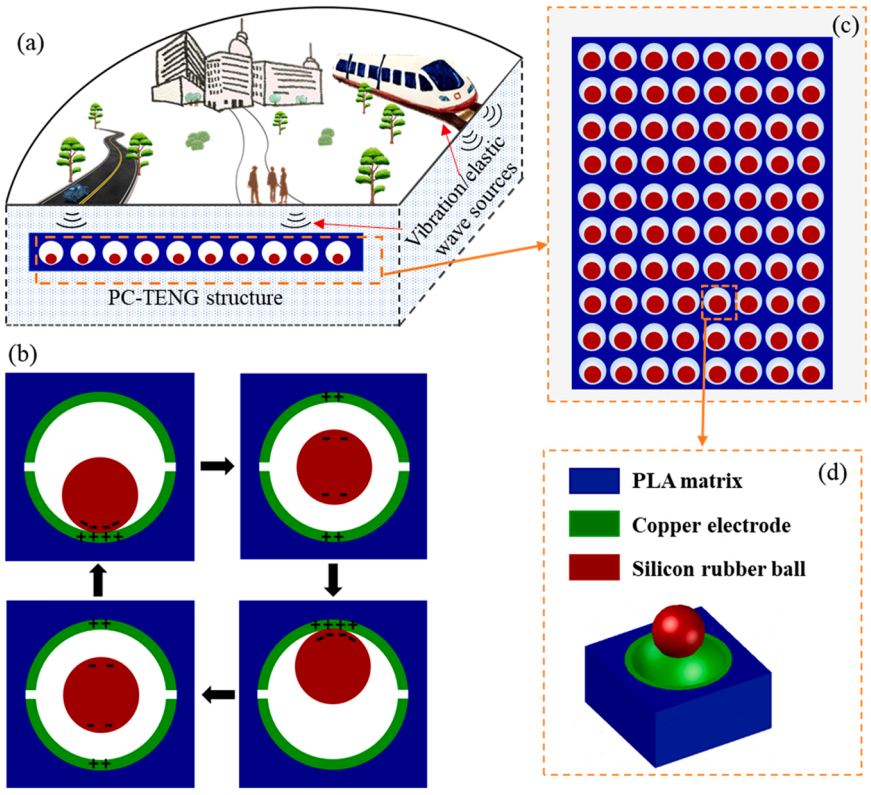


Fig. 1. Schematic diagram of the proposed PC-TENG structure, (a) used in the fields of some smart buildings or transportation systems, (b) working principle of TENG with free-standing layer model, (c) plane graph of the proposed two-dimensional TENG structure, (d), structure of a TENG element.

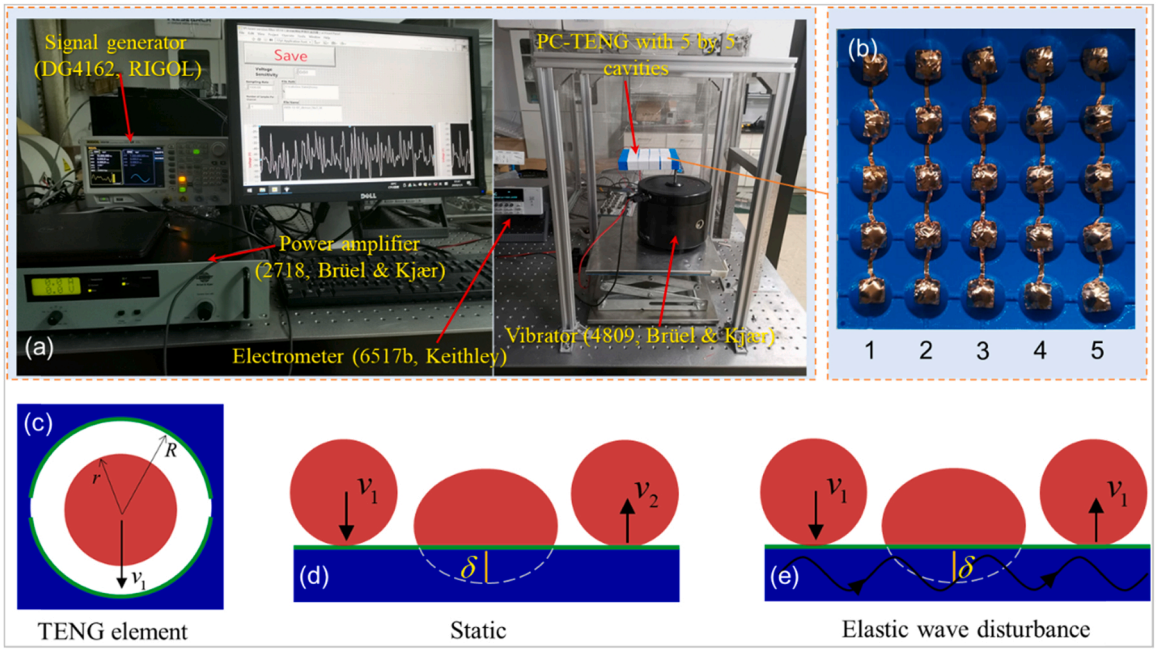


Fig. 2. Experimental setups and sketch diagrams of interaction between the elastic silicon rubber ball and the base, (a) experimental setups used in the experiment, (b) the half part of the fabricated PC-TENG, (c) the geometrical graph of a TENG element, (d) the elastic ball colliding with a static base, (e) the elastic ball colliding with a base under an elastic wave disturbance.

output performance. The Hybrid energy harvesting systems based on triboelectric nanogenerators was provide in [41]. A metamaterial-based TENG for energy harvesting and vibration control was proposed in [42]. Particularly, TENG has been explored in the new fields of in very recent [43–46]. The reader is referred to the review papers [47,48] for details.

In this paper, we propose a two dimensional phononic crystal TENG (PC-TENG) sensor, placed near underground foundation of buildings and transportation systems, monitoring mechanical disturbances to the underground foundation. The PC-TENG is composed of spherical cavities, in each of which there is an identical elastic ball, periodically

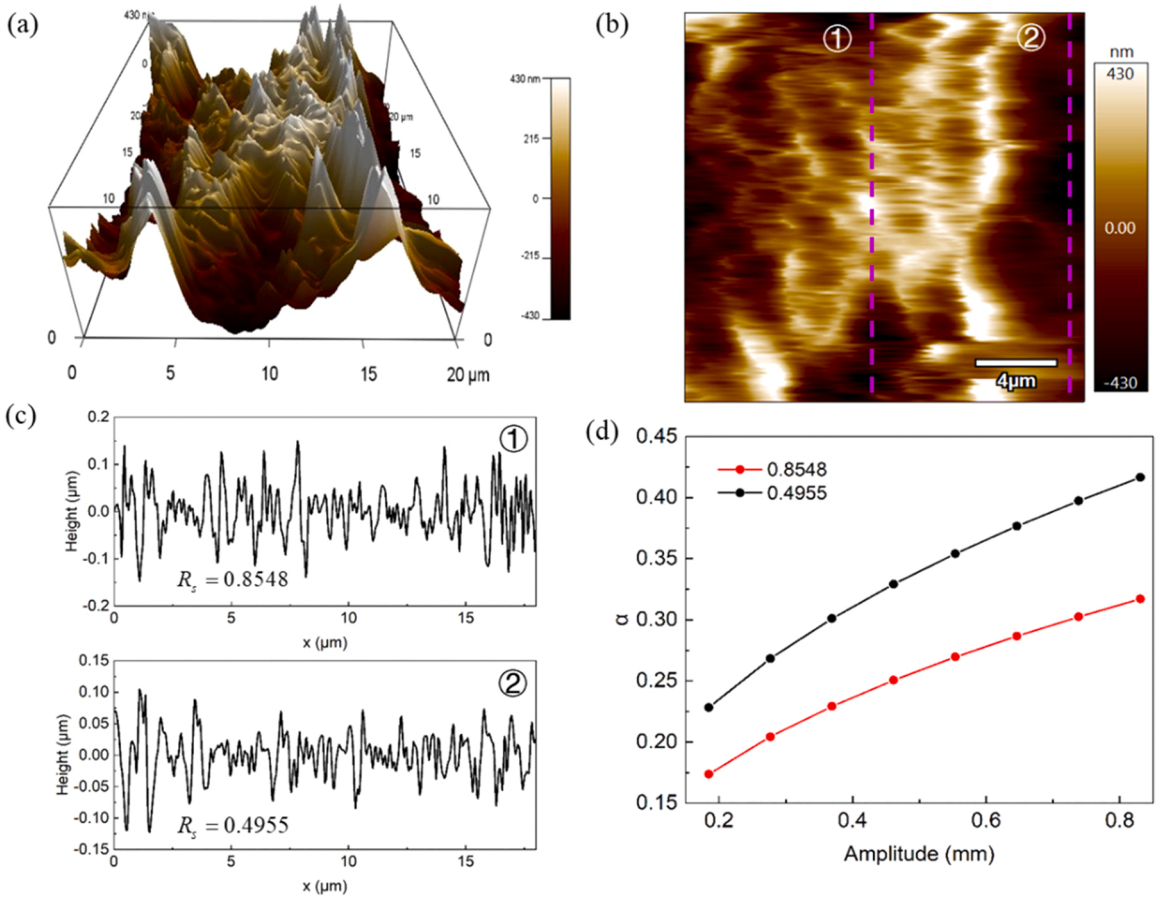


Fig. 3. 3D and 2D surface topographies of the used silicon rubber ball's surface (20 μm×20 μm) by AFM in the experiment, (a) 3D surface topography, (b) 2D surface topography, (c) surface height curves of (1) and (2) located in (b), (d) the ratio of nominal area to real area under different amplitudes when the exciting frequency is 75 Hz and the silicon rubber ball's dimeter is 8 mm. The root mean square slope R_s is obtained from 2D surface topographical figure through Gwyddion software.

embedded in a plate-type of structure as shown in Fig. 1**(c)**. The top and bottom surfaces of each cavity are deposited on (or adhered to) electrode materials. Each cavity unit constitutes a TENG unit when the elastic ball has a movement. It works with a free-standing layer mode, which has advantages of simple structure, easy making, and large output performance. Fig. 1(d) shows the half of each cavity unit. Such a PC-TENG system can be placed in or as the foundation of buildings and engineering equipment. Under some disturbances such as vibration or elastic wave sources resulting from walking people, moving vehicles, landing aircraft to name but a few, the elastic ball moves in each cavity and thus the TENG element works. For the proposed PC-TENG structure, the free-standing mode has advantages of simple structure, easy making, large output performance." As a result, the state of such a disturbance can be detected by the PC-TENG system through the induced electrification charges, similarly, the associated mechanical energy can also be collected through the PC-TENG. Particularly, we can also take advantage of bandgap property of the proposed PC-TENG system to absorb or suppress vibration impact within certain frequencies, and then to protect buildings and equipment from damage. As the proposed TENG system is of sensing and harvesting characteristics as well as bandgap property, it will have a huge potential application in the field of green buildings and intelligent transportation systems. It should be noted that the frequency of disturbances acting on the PC-TENG structure maybe from a few to hundreds (even kilos) hertz, which depends on a variety of factors such as the state of moving objects on roads and surrounding environments. In addition, we only concern the elastic ball bounces up and down under some elastic wave disturbances, other movement modes of the elastic ball are neglected in this paper. In this case, the working principle of the proposed PC-TENG belongs to the freestanding layer mode as shown in Fig. 1(b).

2. Results and discussion

2.1. Theoretical model

For the proposed PC-TENG system, each cavity forms a TENG element as shown Fig. 2(c). Under an elastic wave or a vibration disturbance, the elastic ball bounces up and down and is then periodically in contact with the top and bottom electrodes in the cavity. During such a process, there appears induced triboelectric charges Q, which can be used to sense the state of disturbance and collect mechanical energy as well. We establish a theoretical model incorporating the effects of diameter, surface roughness, and deformation of the elastic ball, for the cavity-ball type of TENG element.

For convenience, assume the surface of each cavity has a displacement $X_b = A \sin \omega t$ due to the elastic wave or vibration disturbance, $\omega = 2\pi f$ and A are circular frequency and amplitude, respectively. Under continuous disturbance, the elastic ball tends to a stable motion in the cavity, namely, the velocity of the elastic ball is uniform, as shown from the measured electrical signals below in Figs. (4), (5) and (6). When an elastic ball collides with a static solid surface, the loss of mechanical energy leads to the bouncing velocity $v_2 = ev_1$ smaller than the incident velocity v_1, see Fig. 2(d), here, e is the coefficient of restitution for the elastic ball colliding with the static base. During the colliding process, the energy loss is

$$\Delta E = \frac{1 - e^2}{2} m v_1^2. \tag{1}$$

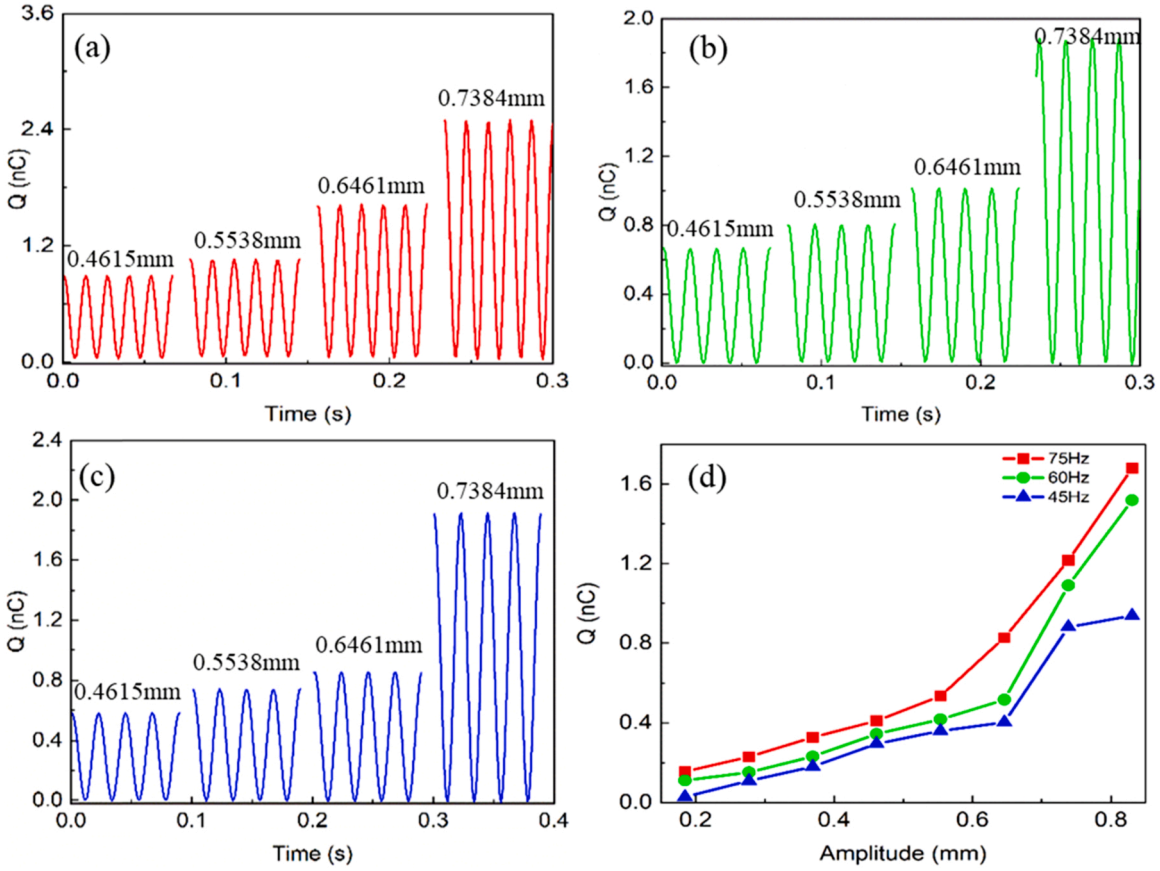


Fig. 4. The output charge of the PC-TENG under different excitations, (a) f= 45 Hz, (b) f= 60 Hz, (c) f= 75 Hz, (d) the plus peak charge under different excitations.

Actually, the elastic wave disturbance as shown in Fig. 2(e) supplies mechanical energy to the elastic ball and thus makes it have a stable motion in the cavity. In this case, the mechanical energy W_e supplied by the elastic wave disturbance should equate the energy loss due to colliding, that is,

$$W_e = \Delta E. \tag{2}$$

In addition, the elastic ball undergoes an elastic deformation during contact process between the elastic ball and the base. We use E and μ, respectively, to denote the Young's modulus and Poisson's ratio of the elastic ball. As the proposed PC-TENG system being one part of building structures, it should be hard enough to ensure safety of structures. In this sense, we regard the base of the PC-TENG as a rigid body. According to the Hertzian contact model [40], the quantitative relation between contact force (T) and depth (δ) of the elastic ball can be written as

$$T = k\delta^{3/2}, \tag{3}$$

where $k = \frac{4}{3}\sqrt{\frac{Rr}{R-r}}\,\frac{E}{1-\mu^2}$. Eq. (3) indicates the contact process between the elastic ball and the base behaves like a nonlinear spring-mass system, k represents the nonlinear spring constant. For the maximum contact depth $\delta_{\max}$, we have the maximum contact force $T_{\max} = k\delta_{\max}^{3/2}$.

The velocity of the base V_b can be determined with the time derivative of X_b, that is, $V_b = \omega A\cos\omega t$. Actually, the real contact process continues a very small time Δt, the velocity of the base during the contact process keeps approximately a constant, we write it as $V_b = \omega A\beta$. Here, β represents the value of $\cos\omega t_0$ at the moment t_0 when the ball comes into contact with the base. Thus, the work W_e done by the base to the elastic ball can be evaluated by

$$W_e = \int_{\Delta t} T(\delta)\cdot V_b dt \approx A\omega\beta\int_{\Delta t} T(\delta)dt = 2Amv_1\omega\beta. \tag{4}$$

For the last equation of Eq. (4), the impulse-moment theorem, that is, $\int_{\Delta t} T(\delta)dt = 2mv_1$ is used. From Eqs. (1), (2), and (4), we can solve the ball's velocity $v_1 = \frac{4A\beta\omega}{1-e^2}$. For the process of the contact depth from maximum to zero, the elastic ball's velocity is from 0 to v_1. In view of the conservation of energy, the maximum elastic potential energy $\int_0^{\delta_{\max}} T dx = \frac{2}{5}k\delta_{\max}^{\frac{5}{2}}$ of the elastic ball is identical with its maximum kinetic energy $\frac{1}{2}mv_1^2$. Immediately, we obtain the maximum contact depth and force from Eq. (3). They are

$$\delta_{\max} = \left(\frac{5}{4}\frac{m}{k}v_1^2\right)^{\frac{2}{5}}, \quad T_{\max} = \left(\frac{5}{4}\right)^{\frac{3}{5}}k^{\frac{2}{5}}m^{\frac{3}{5}}v_1^{\frac{6}{5}}. \tag{5}$$

Then, according to Hertzian contact model [49], we can obtain the maximum contact radius $a_{\max} = \left(\frac{3}{4}\frac{Rr}{R-r}\frac{T_{\max}}{E'}\right)^{\frac{1}{3}}$.

The TENG's real triboelectric charges are proportional to real contact area, which is dependent on contact radius and surface roughness. In this paper, we employ the theoretical model for the elastic contact on a rough surface, developed by Bush and Gibson [50], to evaluate the real contact area between the silicone rubber ball and the copper electrodes. As the surface of cooper electrode is much smoother than that of the rubber elastic ball, we neglect the surface roughness of cooper electrodes for simplification. For the elastic silicon rubber ball used in our experiment, Fig. 3(a) and 3(b) show the 3D and 2D surface topographies scanned by AFM, respectively. For two surfaces, according to Ref. [50], there is the following differential relation

$$dS_r = \frac{qdS_n}{E'}\sqrt{\frac{\pi}{R_s}}, \tag{6}$$

where S_r and S_n denote the real and nominal, respectively, contact area

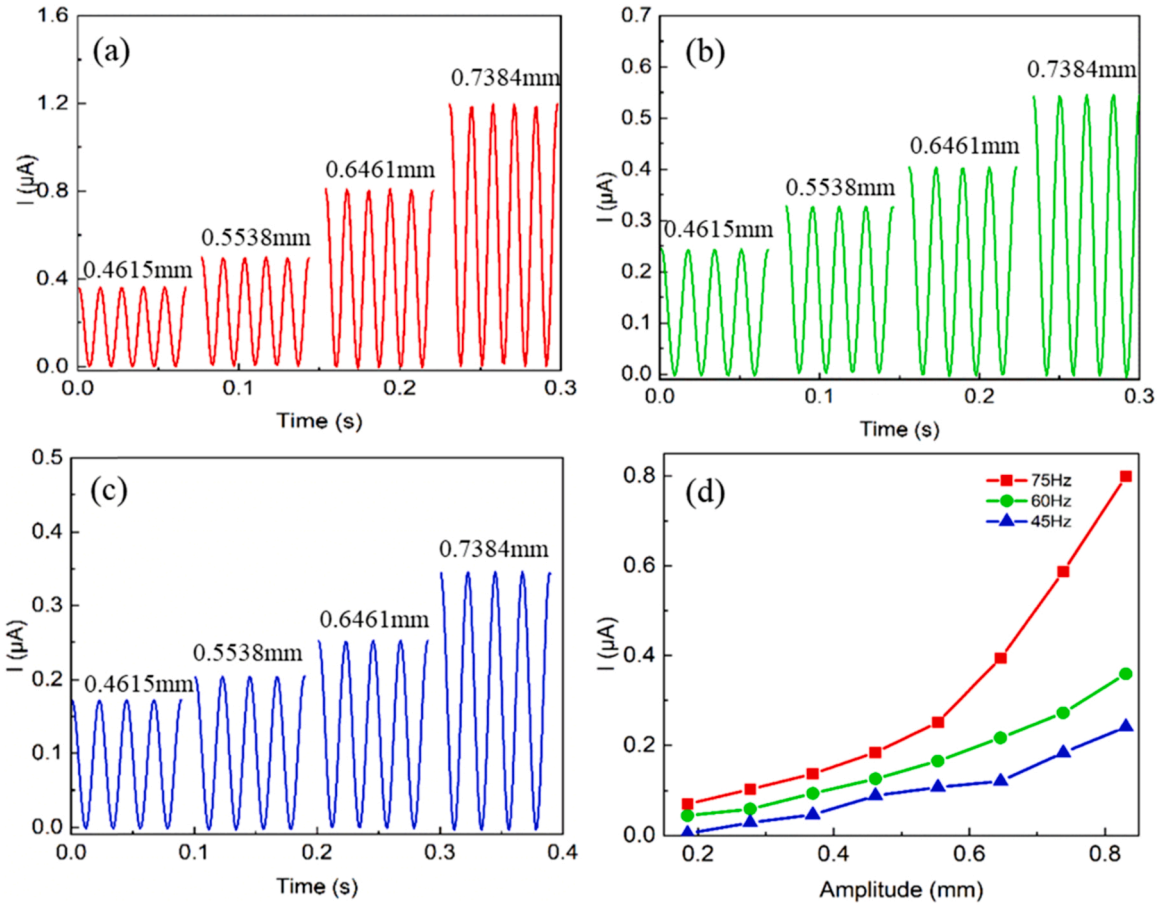


Fig. 5. The output current of the PC-TENG under different excitations, (a) f= 45 Hz, (b) f= 60 Hz, (c) f= 75 Hz, (d) the plus peak current under different excitations.

per unit nominal area, $q = \frac{3T_{\max}}{2\pi a^3}\sqrt{a^2 - r^2}$ is the maximum contact stress, E' the equivalent Young's modulus. R_s is one parameter representing the degree of surface roughness and given [50] by $R_s = \frac{1}{N}\sum_{n=1}^{N}\left(\frac{dz}{dx}\right)_n^2$, N is the total number of data points on the surface profile scan, z is the surface height relative to its mean. From Eq. (6), we can obtain the real contact area

$$S_r = \int \frac{q}{E'}\sqrt{\frac{\pi}{R_s}}dS_n = 12\frac{m^{\frac{3}{5}}}{\sqrt{R_s}}\left(\frac{1-\mu^2}{E}\right)^{\frac{3}{5}}\left(\frac{Rr}{R-r}\right)^{\frac{1}{5}}\left(\frac{A\beta\omega}{1-e^2}\right)^{\frac{6}{5}} \tag{7}$$

Noted that, as shown in Fig. 3(a), for different reference lines,R_s are different. Considering the contact area of silicon rubber ball do not remain consistent in different collisions, we selected two lines labeled (1) and (2) for example in Fig. 3(b), whose surface height curves are plotted in Fig. 3(c), the value of R_s are about 0.8548 and 0.4955, respectively, and used in this paper. Theoretically speaking, the larger the diameter of the ball, the larger the contact area. Define the ratio $\alpha = S_r/S_n$. For the silicon rubber ball with d= 8 mm, with Eq. (7), we can obtain the ratio α under different amplitudes when the exiting frequency keeps 75 Hz, which is plotted in Fig. 3(d). It can be seen from Fig. 3(d) that the real contact area has a rise with the increase of exciting amplitude.

The working principle of the proposed PC-TENG belongs to the freestanding layer mode. The peak output charge Q_m and current I_m are

$$Q_m = \sigma_r S_r = N_Q \rho^{\frac{3}{5}}\frac{\sigma_r}{\sqrt{R_s}}\left(\frac{1-\mu^2}{E}\right)^{\frac{3}{5}}\left(\frac{R}{R-r}\right)^{\frac{1}{5}}\left(\frac{\beta}{1-e^2}\right)^{\frac{6}{5}}A^{\frac{6}{5}}f^{\frac{6}{5}}r^2,$$
$$I_m = Q_m\omega = N_I \rho^{\frac{3}{5}}\frac{\sigma_r}{\sqrt{R_s}}\left(\frac{1-\mu^2}{E}\right)^{\frac{3}{5}}\left(\frac{R}{R-r}\right)^{\frac{1}{5}}\left(\frac{\beta}{1-e^2}\right)^{\frac{6}{5}}A^{\frac{6}{5}}f^{\frac{11}{5}}r^2, \tag{8}$$

where σ_r is the saturation charge density per unit area of certain solids, $N_Q = \frac{1}{3}2^{4.4}\cdot 5^{0.6}\pi^{2.3} \simeq 257.2$ and $N_I = \frac{1}{3}2^{5.4}\cdot 5^{0.6}\cdot\pi^{3.3} \simeq 1615.9$. Eq. (8) shows that the output charges and current are functions of the elastic ball's dimeter, material property and surface roughness, frequency and amplitude of excitations. Hence, the PC-TENG can work as a sensor.

2.2. Sensing performance

Next, we experimentally investigate the sensing performance of the fabricated PC-TENG with 5 by 5 cavities, as shown in Fig. 2(a). For the silicon rubber ball with diameter 8 mm, we measure the output charge Q, current I, and voltage V of the channel 5 near the excited side of the fabricated PC-TENG under different excitations. For the fabricated PC-TENG structure, the frequency response curve of displacement is shown in Fig. 11 below, we choose three typical excitation frequencies of f= 45 Hz, 60 Hz, and f= 75 Hz, at which the frequency response value of displacement reaches about its plus peak, zero, and minus peak,

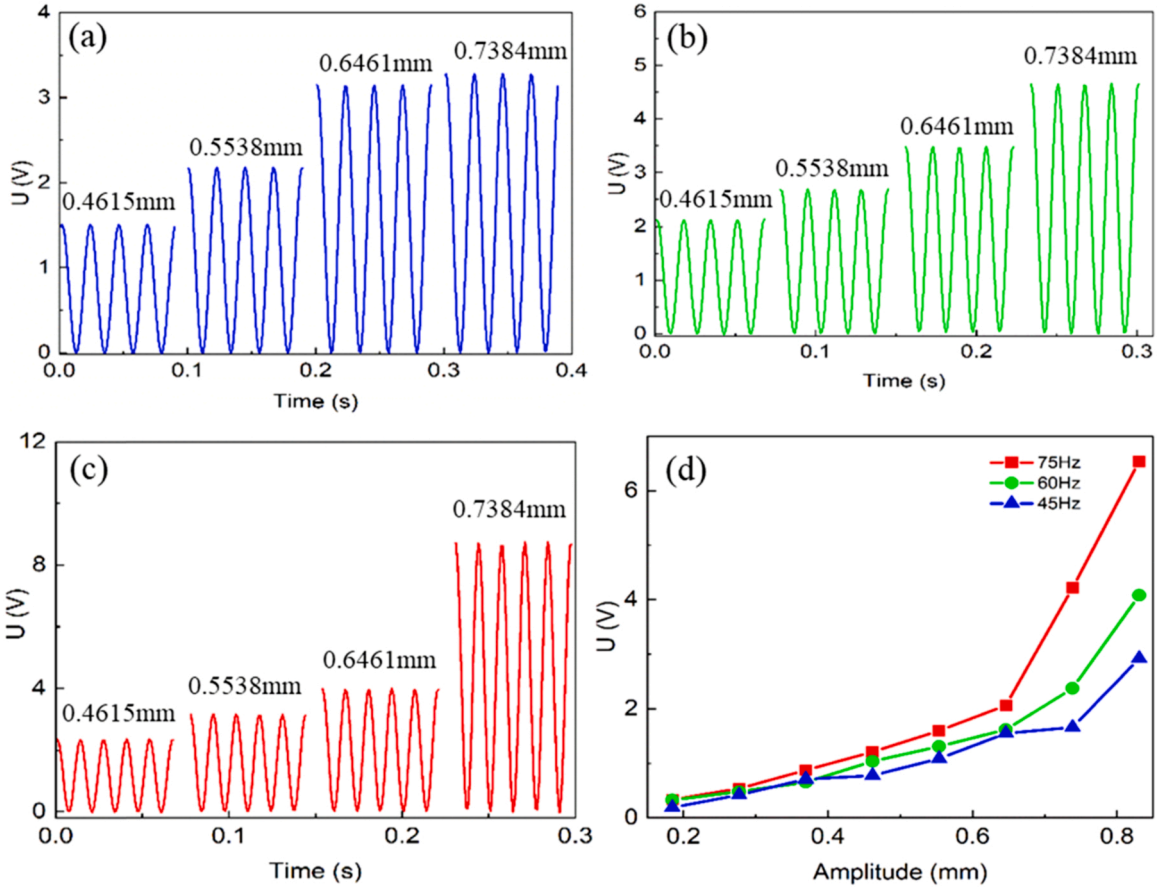


Fig. 6. The output voltage of the PC-TENG under different excitations, (a) f= 45 Hz, (b) f= 60 Hz, (c) f= 75 Hz, (d) the plus peak voltage under different excitations.

respectively. Figs. (4), (5), and (6) show the output signals of Q, I, and V, respectively, of the fabricated PC-TENG under different exciting amplitudes. From Figs. 4(d), 5(d), and 6(d), we find that the higher the exciting frequency f (or amplitude A), the larger the output (voltage, charge, current) signals under the same amplitude A (or frequency f); there are also good mapping relationships between the measured signals and the exciting frequency/amplitude, which indicates the proposed PC-TENG is suitable for sensing the state of an elastic wave or vibration disturbance. In addition, we measure the output signals of Q, I, and V for the exciting frequency ranging from 60 Hz to 80 Hz when fixed the exciting amplitude A= 0.3692 mm, and plot them in Fig. 7, from which we can find that the amplitude of the output signals has a rise with increasing the exciting frequency.

Eq. (8) clearly shows that the elastic ball's diameter is also one key designing parameter for the proposed PC-TENG system. Fixing the exciting frequency (f=75 Hz) and amplitude (A=0.924 mm), we measure the output charge, current, and voltage of the PC-TENG for the elastic silicon rubber ball with three different diameters (d=6 mm, 8 mm, 10 mm), and plot them in Fig. 8. As expected, the output signals for the ball with d= 10 mm are larger than those for d= 6 mm and 8 mm. It can be seen, from Fig. 8, that the output voltage is 15 V when d= 10 mm, which is about 9 times as many as that (1.7 V) when d= 6 mm.

Further, we make a verification of the theoretical model developed in this paper with the experimental data. As R_s and β can't be known exactly, we chose three typical cases, namely, Case 1: R_s= 0.8548, β= 1; Case 2: R_s= 0.4955, β= 1; Case 3: R_s= 0.8548, β= 0.8. In our experiment, the restoring coefficient e is 0.799 which is determined experimentally. The saturation charge density σ_r of the elastic silicon rubber ball is from Ref. [51]. For different diameters of the elastic ball under different exciting frequencies and amplitudes, both experimental data and theoretical results calculated with Eq. (8) are plotted in Fig. 9. It can be seen that, for the theoretical and experimental results, the rising trends of the output charge and current with increasing the amplitude and frequency of disturbances and the elastic ball's radius are agreed well. For the difference between the theoretical and experimental results, one reason is the approximated values of three physical parameters of saturation, restoring coefficient, and surface roughness are used in our theoretical calculations. In addition, the elastic ball will undergo a larger deformation for the larger amplitude and radius and higher frequency. In this case, the derived theoretical model based on the Hertz contact model is failure and thus leads to a larger difference.

Finally, we measure the output charge, current, and voltage of each channel, see Fig. 2(b), simultaneously for the fabricated PC-TENG under two different excitations (f=45 Hz, A=0.476 mm) and (f=75 Hz, A=0.1846 mm), respectively, and plotted them in Fig. 10. The elastic silicon rubber ball is with diameter d= 8 mm. It can be seen that the variation of the measured signals has an opposite trend (from Chanel 1 to Chanel 5) for two different exciting frequencies (f=45 Hz and 75 Hz). This is because the fabricated PC-TENG belongs to a phononic structure

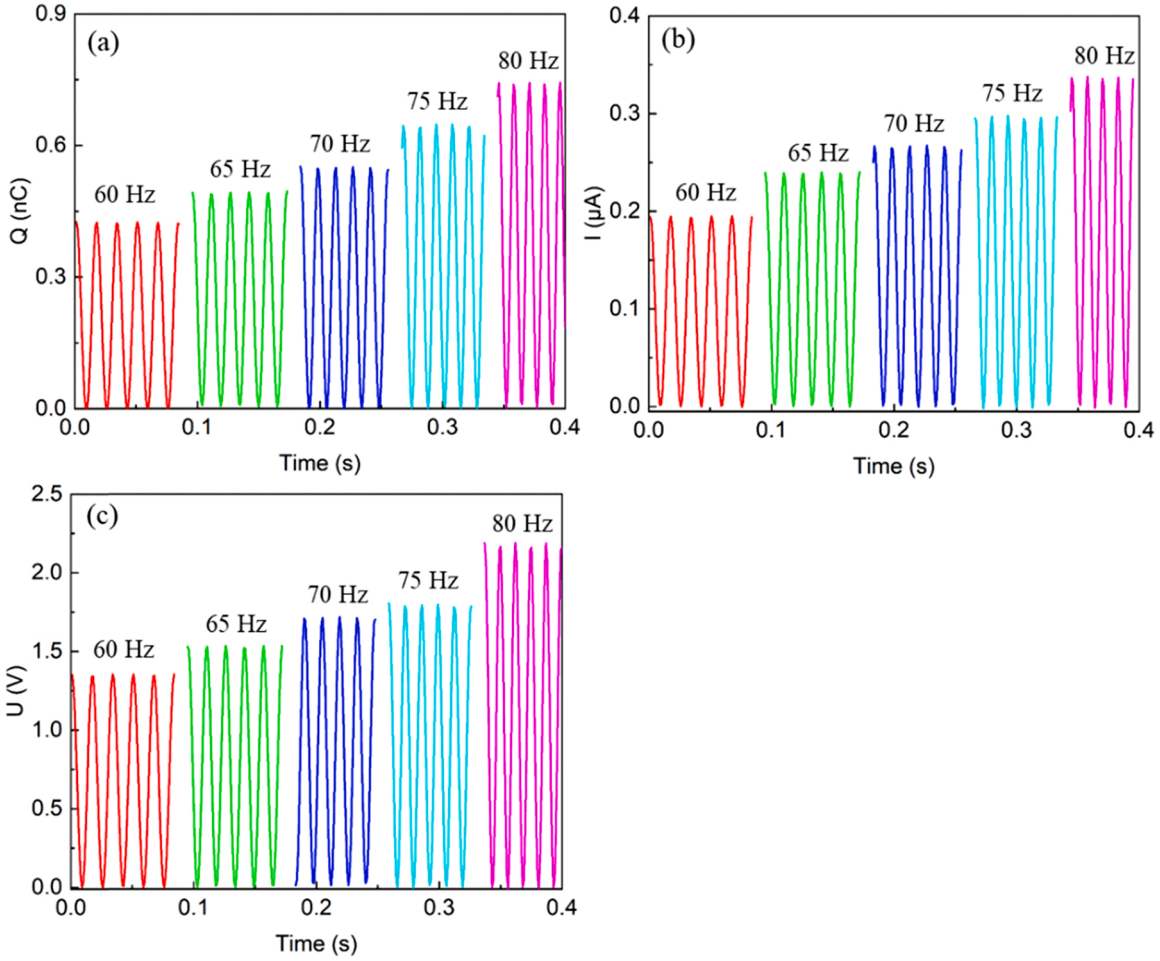


Fig. 7. The output signals of PC-TENG under different excitations. (a) charge, (b) current, and (c) voltage under different exciting frequencies from 60 Hz to 80 Hz.

with typical bandgap characteristics. Fig. 11 is the measured frequency response curve of displacement of the fabricated PC-TENG structure, which shows that the exciting frequencies of 45 Hz and 75 Hz are within passband and forbidden gap, respectively. For f= 45 Hz and 75 Hz, the total output power of the fabricated PC-TENG are about 0.107μW and 0.105μW, respectively. For the fabricated PC-TENG, the ratio of the harvesting efficiency under f= 75 Hz (forbidden gap frequency) to that under f= 45 Hz (passband frequency) is about 2.57, which indicates that the proposed PC-TENG system has a better harvesting performance if the exciting frequency is within forbidden gaps. Hence, the PC-TENG system proposed in this paper can efficiently collect mechanical energy but also work as a vibration and shock absorber for some major buildings and equipment.

3. Conclusion

We propose a phononic crystal-based TENG sensor, which can be used to monitor the state of elastic wave and vibration disturbances. The proposed PC-TENG consists of a solid base embedded with periodic spherical cavities and elastic balls. Based on the Hertizan contact model, we present a theoretical model, incorporating the elastic ball's diameter, material property, surface roughness, and the exciting frequency and amplitude of disturbances, for the proposed PC-TENG. For the fabricated PC-TENG with 5 by 5 cavities, composed of the PLA base and elastic silicon rubber balls, the measured output charge, current, voltage for different excitations show that the PC-TENG has a good sensing performance. The developed theoretical model is verified with the experimental data. The results show the proposed PC-TENG can sense amplitude and frequency of an elastic wave disturbances and can be also used to absorb and suppress shock impact within certain frequencies from surrounding disturbances. In all, this paper provides the concept of a phononic crystal-based TENG system for smart buildings and transportation systems. However, for the PC-TENG sensor there are more issues, such as experimentally demonstrating the effect of surface roughness on its sensing performance and the durability of the PC-TENG sensor, should be studied further.

4. Experiment methods

The PC-TENG base with 5 by 5 cavities used in our experiment is fabricated by a 3D printing machine with polylactic acid (PLA) materials. The filling rate is 20%. Fig. 2**(c)** shows the half part of the PC-TENG base, the other part is the same as it. The fabricated PC-TENG structure has 25 (5 × 5) spherical cavities with diameter 16 mm, the spacing of

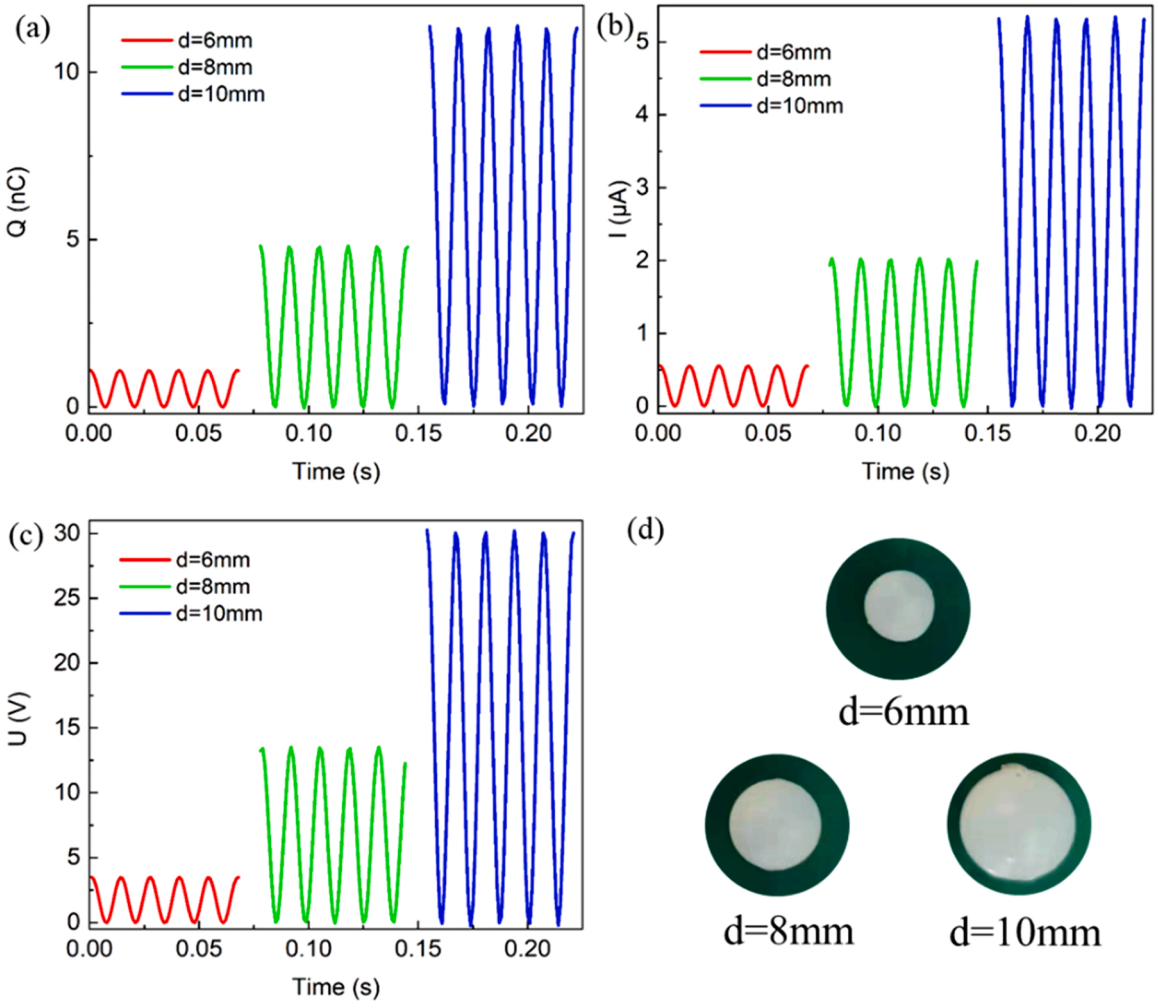


Fig. 8. The output signals of the PC-TENG under the excitation with f= 75 Hz and A= 0.924 mm, (a) charge, (b) current, (c) voltage, (d) three elastic rubber balls with diameters of 6 mm, 8 mm and 10 mm.

centers of two adjacent spherical cavities is 20 mm, and the total thickness of the PC-TENG is 20 mm. Here, spherical top and bottom surfaces of each cavity are partly adhered with copper electrodes. An elastic silicon rubber ball is placed in each cavity. Thus, it forms one TENG element for each cavity. The 25 cavities are divided into 5 channels, which are labeled by1–5, respectively, from the left side to the right side, see Fig. 2(b), the five electrodes in each channel are connected in series. For each experimental test, the same PC-TENG base and copper electrodes are used.

The relevant instruments for measuring electrical signals mainly include vibrator (4809, Brüel & Kjær), signal generator (DG4162, RIGOL), power amplifier (2718, Brüel & Kjær) and electrometer (6517b, Keithley) as shown in Fig. 2(a). The sinusoidal signal of a given frequency can be set in the signal generator. The amplitude of the sinusoidal signal can be adjusted by the power amplifier, and the amplifier would output the sinusoidal excitation of a given amplitude and frequency to the vibrator. In our experiment, the fabricated PC-TENG is excited on the left side by vibrator with certain exciting frequency and amplitude. Under such an excitation, there will a shear wave propagating in the fabricated PC-TENG base, which will lead to the elastic ball in each cavity bouncing up and down. For TENGs, the accurate measurement of output voltage has been discussed in [52,53]. In our experiments, we adopt the non-grounded method to measure the open-circuit output voltage with electrometer (6517b, Keithley), the internal resistance of electrometer is 200 TΩ. For the output charge and current, we measure them under the electrically shorted condition.

CRediT authorship contribution statement

Lifeng Zhu: Methodology, Investigation, **Zhicheng Zhang**: Supervision, Conceptualization, Writing – original draft, **Dejuan Kong:** Investigation, Visualization, Data curation, **Chengbin Liu:** Investigation, Visualization, Data curation, **Zhigang Cao:** Investigation, Visualization, Data curation, **Weiqiu Chen:** Writing – review & editing, **Chunli Zhang** Supervision, Conceptualization, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

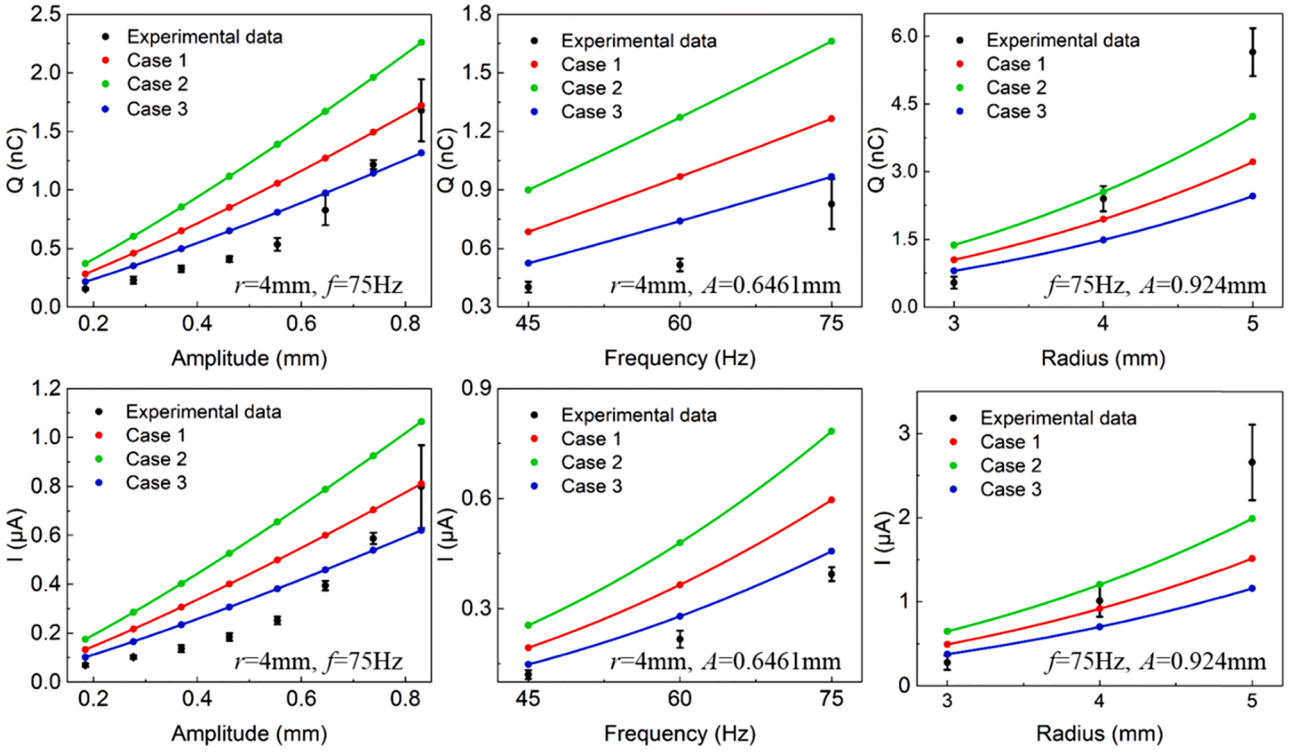


Fig. 9. The measured output charges and currents of the fabricated PC-TENG under different excitations.

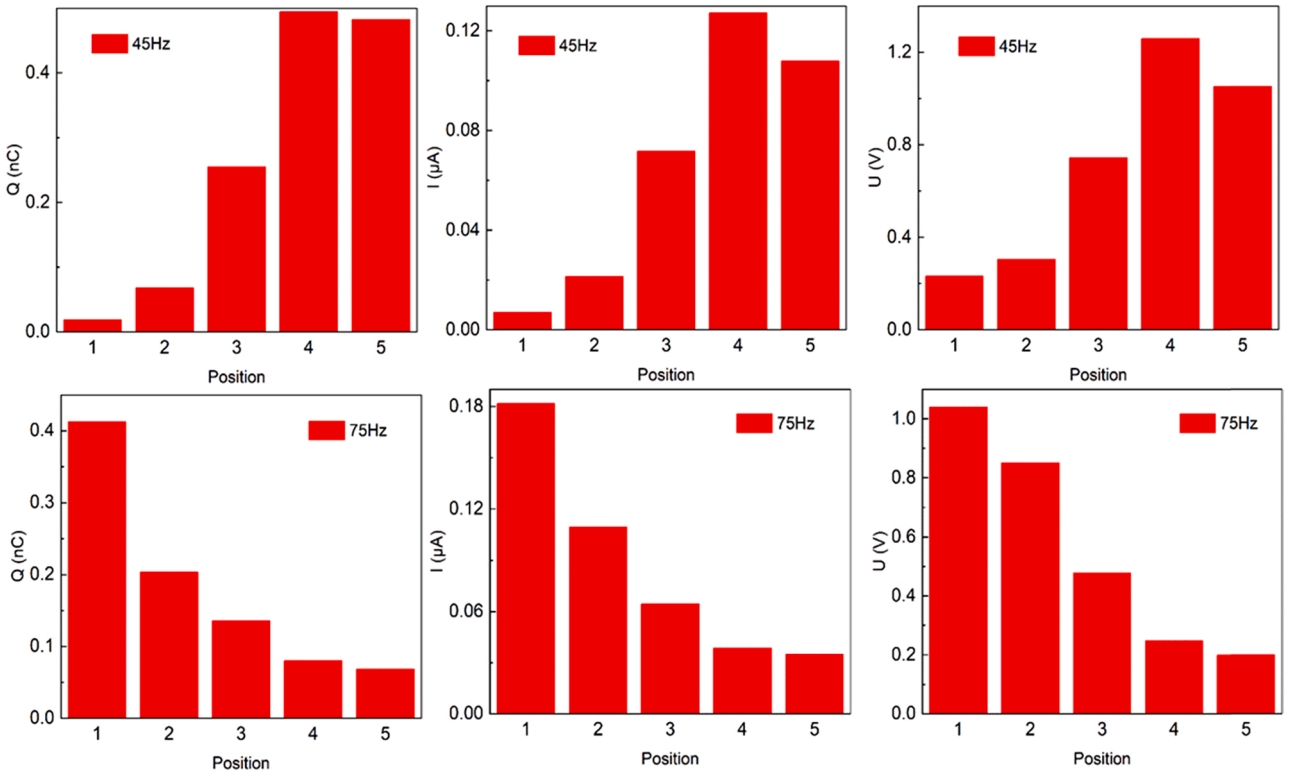


Fig. 10. The measured signals in each channel when the fabricated PC-TENG is under excitations of (f=45 Hz, A=0.476 mm) and (f=75 Hz, A=0.1846 mm), f= 45 Hz and 75 Hz corresponding to the passband and forbidden gap frequency, respectively.

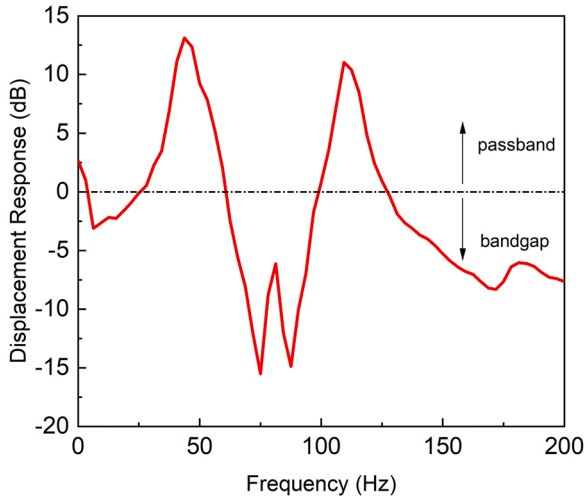


Fig. 11. Frequency response curve of the PC-TENG structure.

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